

1.5W Power Amplifier

RAJ K. GORKHALI

Here we put all the theory to work and present a simple power amplifier module that can be easily built with readily available components. The block diagram of the amplifier is shown in Fig. 1. It is typical of most audio amplifiers, although the circuit is somewhat different.

A power amplifier contains audio input, amplifier, driver, output and power supply sections. The amplifier section provides most of the voltage gain. The driver stage is a buffer between the amplifier section and the output stage. The output stage usually has to drive a low-impedance load such as a loudspeaker. The power comes from the power supply, and the output signal appearing across

the load should ideally be a replica of the input signal. In other words, the output stage takes power from a DC supply to boost the signal so it can drive a load.

The circuit shown in Fig. 2 shows the amplifier, driver and output sections. The amplifier section is built around JFET VHF/UHF amplifier 2N5484 (T1) and npn transistor BC548 (T2). The driver section is built around transistor BC639 (T3) while the output section is built around transistors BD139 and BD140 (T4 and T5).

The input signal is coupled to volume control VR1 via capacitor C1. The value of VR1 is specified as 1-mega-ohm. Since the gate terminal of FET (T1) can be regarded as an open circuit, the input impedance of the circuit is equal to the value of VR1. Like all audio volume controls, VR1 needs to have a logarithmic taper (usually denoted as 'type C') to give an apparent linear relationship between rotation of the control and the volume level. This is necessary because human hearing follows a logarithmic response, in which a change in the output power by

a factor of 10 is heard as a change by a factor of two.

The FET stage in amplifier section is used to give a high input impedance. The next stage is common-emitter amplifier comprising transistor T2. Preset VR2 is used to adjust amplification and avoid direct coupling between transistor stages T2 through T5. This means that DC voltages for T3, T4 and T5 are all determined by the collector voltage at T2. The most important voltage is at the emitters of T4 and T5. Preset VR2 is used to adjust this to half the supply voltage. To stabilise this and other voltages in the circuit, resistor R13 gives negative feedback from the output to the emitter of transistor T2. If capacitor C8 is not included, the feedback will be for both DC and AC voltages. It will be for DC only if C8 is added.

When voltage at the emitters of transistors T4 and T5 rises, say, due to a temperature change, the voltage at the emitter of transistor T2 will also increase by way of R13. This will cause T2 to conduct less current, making the DC voltage at its collector increase. As

a result, transistor T3 will conduct more current and its collector voltage will drop. This then reduces the voltages at the bases of T4 and T5 and hence their emitter voltages.

Driver transistor T3 and its collector load is the base circuitry associated with T4 and T5. In effect, T3 is connected as a common-emitter amplifier. The output signal developed across T3 is applied to the base of T4 via

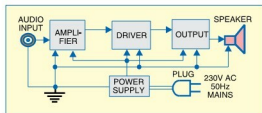


Fig. 1: Block diagram of 1.5W power amplifier

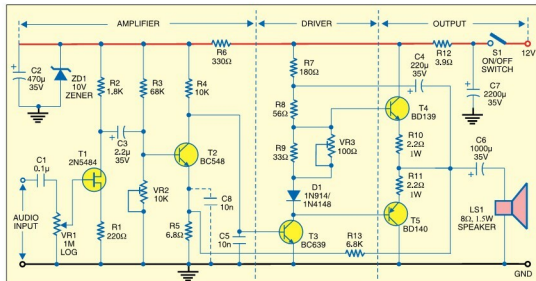


Fig. 2: 1.5W power amplifier circuit

